# ReGainX: ML-Based Exoskeletons for PD Motor Impairments

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## Introduction

### Parkinson's Motor Impairment

- ★ **Parkinson's disease (PD)** impairs over **10 million people**[1], costing the U.S. **$52 billion annually**[4]
  - **Incurable & rising** amongst aging global cases[1]
  - Disrupts muscle activity → **morbidity & mortality**[2]
  - Caused by loss of **dopamine**-producing neurons[3]
- ★ **Bradykinesia & muscular degeneration**
  - Restricts daily activities → **pain & dependency**[2]
  - **Major gap** in symptom monitoring[5]

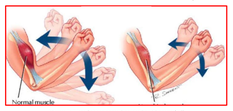


Figure 1: Muscle Degeneration Motion Diagram(16)

#### Muscular Degeneration

- ★ **Occurs across all stages of PD**[6]
  - **Progressively evolves**[6] → solutions must adapt
- ★ **Sarcopenia**[3] & **fatigue**[2] limit patient motion
  - Predominantly **lower-limbs**[7] **& elbows**[10]
- ★ Linked to **mitochondrial dysfunction**[8] (low muscle power) & **lack of physical activity**[2]
- ★ **Potential** of assistive devices in restoration[9]

$$F_{deg} = \alpha_1 \cdot F_{max} \cdot e^{-(\alpha 2+\beta t)} = \text{m_loss} \cdot \text{m_fati}$$

Equation from Myosuite (Caggiano et al.) (18)

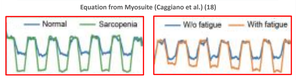


Figure 2-3: Myosuite's Sarcopenia + Fatigue on Muscle Activations (18)

#### Bradykinesia

- ★ Affects **98%** of PD patients[12]
- ★ Significant **slowness** & delay (**akinesia**) of movement, often asymmetrical[14] → **fall risk**
  - **10-40%** slower[13]
  - Mental challenges: **frustration**[17]

$$N_{brady} = x(1-\Delta i) \cdot \alpha = \text{delay \& slowness}$$

Bradykinesia Equation Created by research with reference from (17)

- ★ Caused by **basal ganglia dysfunction**[14]
- ★ Sensors (**IMU**) can assess severity[15]

Figure 4: Simulated PD patient - Created by Researcher

### Limitations

**[1] - Adaptive** nature of RL & randomized simulator
**[2] - Low-cost**, accessible product
**[3] - High-fidelity**, validated simulator
**[4]** - Intelligent **EMG/IMU** analysis for movement & impairment prediction
**[5] - Wide-ranging** set of exoskeletons
**[6] - Cloud Computing Pipeline**

**Implemented in theory, not in practice → lacks real-time & patient-adaptive functionality.**

| | | | |
|---|---|---|---|
| Patient Specific | [1][5] | Expensive | [2][6] |
| Data Overfitting | [1][3] | Manual Control | [4][6] |
| Invasive + Side Effects | [3][4] | Simplified mechanics | [3][4] |

## Research Objective

**Research Question**
**Can machine learning systems achieve real-time, adaptive mitigation and tracking of Parkinson's motor impairments across diverse patients?**

### Criteria

- ★ **Real-time functionality**
  - **Seamless**, natural, effective interaction
- ★ **Adaptive → High restoration %** across all patient PD levels & movements
- ★ **Safe, non-invasive**
- ★ User-friendly → **intuitive** device

### Constraints

- ★ **Real patient data unavailable** → sim-to-real transfer
- ★ **Low-cost** exoskeleton → hardware & model limitations
- ★ Limited **computational resources**
  - Lenovo PC hardware (16GB memory)

### Engineering Goal

**Engineering Goal: Use Reinforcement Learning (RL) to create an adaptive, low-cost solution for motor impairments from Parkinson's Disease**

**Primary Goal**
Provide real-time assistance (**<250ms**) with **over 90%** movement restoration

**Secondary Goal**
Monitor impairment progression with an assistance correlation (r) **above 0.95**

*ReGainX assists across all stages and movement, providing the **first exoskeleton system** tailored to **all** PD patients*

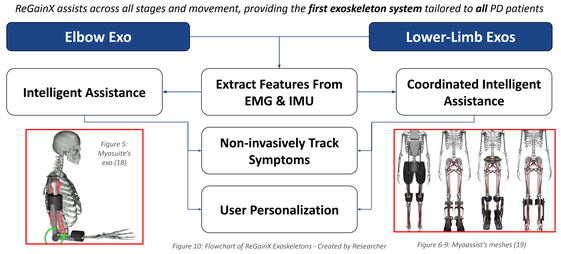


Figure 5: Myosuite's exo (18)

Figure 10: Flowchart of ReGainX Exoskeletons - Created by Researcher

Figure 6-9: Myoassist's meshes (19)

## Engineering Methodology

### Design Approach

- ★ **Reinforcement Learning**
  - Learning through **trial & error**
    - **Markov decision process**
    - Actions → Outcomes → Rewards/Penalties
  - **Superior to traditional AI at tasks humans don't know the answer to**
  - Excels at complex control systems[23]

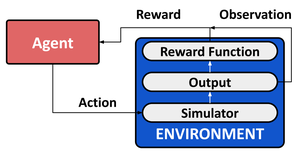


Figure 11: RL Basic Agent-Environment Loop - Created by Researcher

**Alternative Solutions Explored**
Smart-cueing footwear, AI-controlled dosing, VR rehabilitation system, manual exoskeletons, EMG feedback band.

```
action,_ = exo.predict(obs)
obs,reward,done, _ = env.step(action)
```

Figure 12: Markov Decision Process – Created by Researcher

## Engineering Methodology

### 1. Multi-Agent Training Framework

- ★ **Phase 1: Training on Healthy Body**
  - Deep policy successfully performs **randomized tasks** given goals (rewards)
  - Learns **muscle control via neural drive signals**
- ★ **Phase 2: Policy Transfer to PD body**
  - PD modeling → **error in tasks & movement**
- ★ **Phase 3: Multi-agent system:**
  - PD patient + exo agent
  - **Custom reward function** guides learning
    - "Good job, you restored x% motion!"

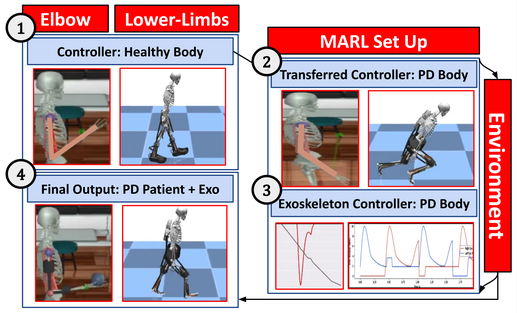


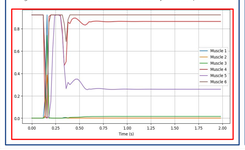
Figure 13: MARL Training Pipeline Flowchart utilizing Myosuite (18) & Myoassist (19) Simulators - Created by Researcher

### 2. Sensor Data

**EMG**

- ★ **Neural drive signals**
  - Derived from EMG:
  - Elbow → **6-m**, Legs → **22-m**
  - Measures **neural drive**[22]
  - **Unnatural** bursts reflect bradykinesia
  - **Partly represents "intention"**

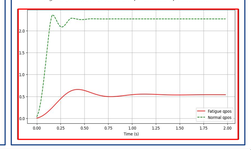
Figure 14: Muscle Excitation: Weak vs. Healthy - Created by Researcher



**IMU**

- ★ **Motion data** - joint kinematics
  - Derived from IMU sensors
    - Joint angle and velocity
  - Muscle degeneration + bradykinesia → **slower & weaker** IMU data
  - **Represents actual motion**

Figure 15: IMU: PD vs. Healthy - Created by Researcher



### 3. Training Environment Construction

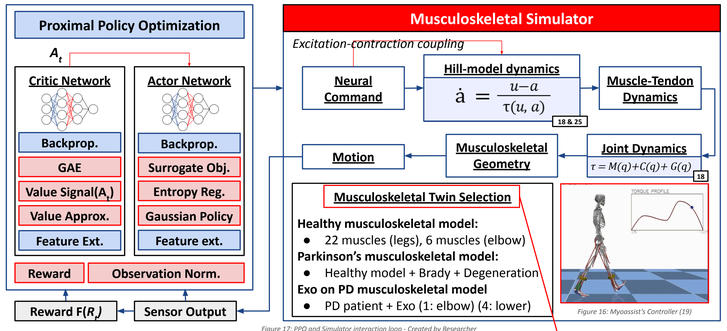
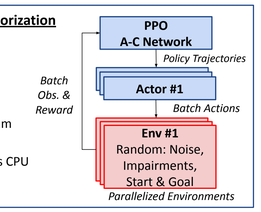


Figure 16: Myoassist's Controller (19)

Figure 17: PPO and Simulator Interaction loop - Created by Researcher

**Environment Vectorization**

- ★ Multiple **random** simulators **run in parallel**
  - Efficient hyperparam tuning
  - Maximizes CPU efficiency



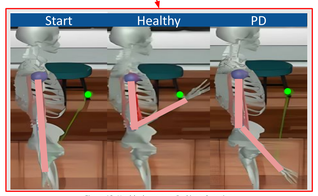



Figure 19: Healthy human vs. Parkinson's patient posing elbow in simulator (18) - Created by Researcher

**Reward Function Design**

$$R_t = -w_1(\Delta q_j)^2 - w_2 \Delta v_j \cdot (\Delta v_j > t) - w_3(\tau_j^{exo})^2 + w_4(\epsilon_t - \epsilon_{t-1})$$

1. **Target position penalty**
   - Reward for lower displacement from goal
   - Calculated relative to previous timestep
2. **Smoothness reward**
   - Penalizes sudden speed changes
3. **Energy efficiency reward**
4. **Restoration reward**
   - Parallel simulation: healthy model on same task compares alongside PD, exo-assisted motion

**Training & Tuning**

- ★ **Hyperparameter Tuning**
  - **Bayesian Optimization**
    - Probabilistic model selects hyperparams
  - **Median pruner**

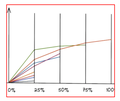


Figure 20: RL Pruner (24)

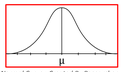
- ★ **Curriculum Learning**
  - **Reward weight tuning** → agent gradually builds upon previous experiences
- ★ Overcomes slow convergence

### 4. Model Configuration

- ★ **Adaptive Variability**
  - 400,000 episodes for training → **400,000 different patient instances**
  - **Domain randomization** → Each episode **randomizes** tasks and severity
- ★ **Critic**
  - **Value Estimation** → long-term reward
  - **Advantage Calculation (GAE)** → how much better the agent's action is than expected

**Value Estimation**
$$V(s_t) = Q(r_t, s_t)$$

- ★ **Stable Learning**
  - **Gaussian Policy** → outputs probability distribution
  - **Clipped Surrogate Objective** → stable policy updates
  - **Entropy Regularization** → exploration vs. exploitation



Normal Curve - Created By Researcher

- ★ **Feature Extractor Architecture**
  - Observation Space: Elbow - **8D** & Lower-limbs - **10-14D**
  - Temporal 2D CNN
    - Dual embedding path → filters capture **global & local** dynamics
    - **Attention to dominant activations** → eliminates noise → **richer obs.**
    - EMG-IMU Fusion → **context-aware control**

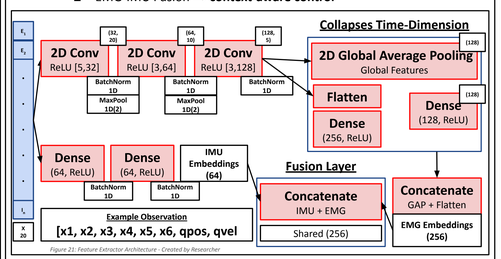


Figure 21: Feature Extractor Architecture – Created by Researcher

**RLHF** - Reinforcement Learning through Human Feedback

- ★ **User feedback interface**
- ★ **Personalization via metadata**
  - System tracks context (time, date, etc.)
  - Shapes device behavior
- ★ **Human-Aligned Learning**
  - Exoskeleton adapts to user's goals.
- ★ **Innovative Application**
  - Unique use of RLHF
  - Producer-user gap

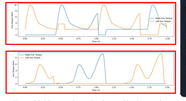


Figure 22-23: N vs. Smooth - Created by Researcher

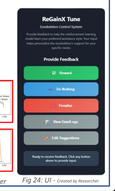


Fig 24: UI - Created by Researcher

## Results

**Symptom Monitoring**

- ★ Exo assistance **correlates** with motor impairment severity
  - **Assistance varies patient-to-patient** → dynamic policy
- ★ Data is automatically logged by exo on ReGainX Tune app
- ★ Current methods: **subjective online tests**[5]

Figure 25: Torque vs PD - Created by Researcher

**Cloud Exoskeleton System**

- ★ **Sim-to-real transfer**
  - RegainX's models run on low-cost hardware, proving **viability**
  - **High-fidelity simulation** eliminates severe model **overfitting** to unrealistic dynamics.
    - Real-time results validate use of deep models in real envs
- ★ **Cloud-based operation** (reference fig 26)
  - Exo outputs torque every **0.1s** (5Hz)
    - Data is logged to **Firebase**
  - **Average cloud latency: ~0.029s** ~35Hz
- ★ **Low-Cost Design** - $130
  - **Arduino**(C++)**, DC driver, and motor**
  - 12V battery → **daily assistance**
  - Easy-to-use straps + lightweight

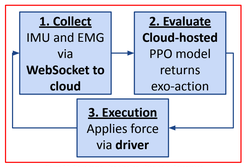


Figure 26: Cloud computing pipeline - Created by Researcher

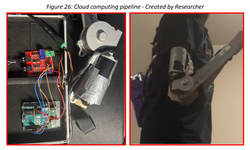


Figure 27: Hardware Pipeline - CBR

Figure 27: Researcher testing Elb-Exo

**Accuracy Matrices**

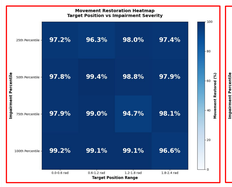


Figure 26: Elbow Exo Heatmap - Created by researcher

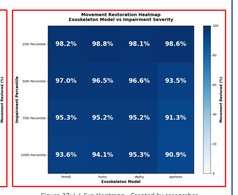


Figure 27: L-L Exo Heatmap - Created by researcher

- ★ **Elbow Exoskeleton**
  - Accuracy = **97.86%**
  - STDEV = **2.72%**
- ★ **Lower-limb Exoskeletons**
  - Accuracy = **95.51%**
  - STDEV = **2.34%**

**Performance Graphs**

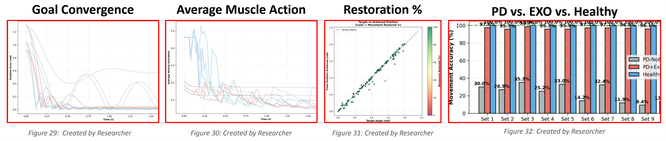


Figure 29: Created by Researcher

Figure 30: Created by Researcher

Figure 31: Created by Researcher

Figure 32: Created by Researcher

**Latency**

- ★ **1.19ms** on ReGainX PPO
- ★ **Lower than frequency**

Figure 28: PPO vs baseline latency- Created by Researcher

**Generalizability**

- ★ **Stratified**, random evaluation methodology: 64,000 each.

## Engineering Methodology

### Ablation Study: ReGainx Model vs. Baseline Model

**Training Performance**

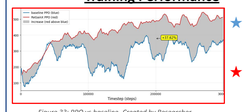


Figure 33: PPO vs baseline- Created by Researcher

- ★ **87.74% accuracy** (↓9.35% vs. ReGainX )
- ★ **37.6%** ↓training performance
- ★ Elementary approach
  - **Ablation Study**
  - Default hyper params
  - Low exploration, **lr**, & rollouts(<100)

Figure 34: Baseline Matrix - CBR

- ★ Base PPO
- ★ RGX PPO

**Temporal Difference Error**

- ★ **Critic's prediction error** for future rewards
- ★ Reduces TDE by **61.4%** → stable predictions
  - STDEV = %0.55
  - **Feature ext. + tuned hyper params**

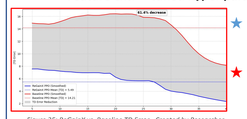


Figure 35: ReGainX vs. Baseline TD Error - Created by Researcher

- ★ Base PPO
- ★ RGX PPO

**TD Err. Equation**
$$T = R_t + \gamma V(s_{t+1}) - V(s_t)$$

$R_t$ = reward at timestep(t)
$V(s_{t+1})$ = future reward *after* $R_t$
$V(s_t)$ = future reward *before* $R_t$

## Conclusion

- ★ This research shows that exoskeletons can be used to **(a)** assess movement and impairment severity, **(b)** mitigate impairment in real-time, and **(c)** monitor PD progression.

### Engineering Criteria

- ★ **Natural, effective real-time functionality**: 100ms assistance intervals & high accuracy
- ★ **Adaptive across all users:** Patient-specific processing & evaluated across varying conditions
- ★ **Safe, non-invasive:** Non-invasive sensors & low-risk product
- ★ **User-friendly:** Intuitive control, easy-to-use velcro straps, user personalization through app

### Key Outcomes

- ★ **Overcomes existing limitations**: low-cost, state-of-the-art exos
  - No fixed assistance strategy → adaptable
- ★ Custom feature extractor to process EMG & IMU streams to guide **prediction of user movement & impairment severity**
  - Utilizes two different tasks to mitigate and track motor impairments
- ★ **Proof of concept:** RL-driven, patient-specific, daily-use tech
  - Lack of heavy computation, patient data, & resources
  - **Scientific value**
- ★ Unique application of RLHF → patient is "expert"

**97.8%** Elbow Movement Restoration

**95.5%** L-L Movement Restoration

**96.9%** Correlation with PD severity

### Impact

- ★ **ReGainX enhances quality of life** for PD patients, primarily those lacking access to medication
  - **Reduced muscular fatigue, delay, & pain** → improves **independence** & daily functioning
  - Reduces morbidity and **mental effects of PD**
- ★ PD severity monitoring → patients & clinicians can monitor progression through app
  - **Eliminates need for costly, time-consuming clinical examination**
- ★ **Accessible** & practical real-world-use
  - **Low-cost**, modular exoskeletons targeted at PD symptom sites.

## Key References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